DISTRIBUTED SYSTEM WITH ACCESS CONTROL INVOLVED

STEP :1 VLAN CREATION

Vlan 10 , 20

SW1:

```
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#
```

SW0(DISTRIBUTED SWITCH IN NETWORK 1):

```
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#
```

SW2:

```
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 30
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#exit
Switch(config)#
```

STEP 2:

VLAN 30 , 40

SW4:

```
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 30
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#exit
Switch(config)#
```

SW3(DISTRIBUTED SWITCH IN NETWORK 2):

```
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 30
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#exit
Switch(config)#
```

SW5:

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 30
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#exit
Switch(config)#
```

SWITCHPORT ACCESS CREATION

STEP:1

VlAN 10 ,20

SW1:

```
Switch#
Switch#en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
```

SW0(DISTRIBUTED SWITCH IN NETWORK 1):

```
Switch#EN
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch(config-if)#
```

SW2:

```
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk
FastEthernet0/1 VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/1 on VLAN0001. Inconsistent port
type.


%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

STEP 2:

VLAN 30 , 40

SW4:

```
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface f0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#
```

SW3:

```
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#interface fa0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch(config-if)#exit
Switch(config)#
```

SW5:

```
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#exit
Switch(config)#%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk
FastEthernet0/3 VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/3 on VLAN0001. Inconsistent port
type.
```

ROUTER CONFIGURTAION

STEP 1:

VLAN 10 , 20

ROUTE 0:

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface giga0/0/0
Router(config-if)#no shutdown
```

```
Router(config-if)#exit
Router(config)#interface giga0/0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.10, changed state to
up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip add 192.168.1.10 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface giga0/0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.20, changed state to
up

Router(config-subif)#ip add 192.168.2.10 255.255.255.0

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL vLAN.

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip add 192.168.2.10 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
```

SW0(DISTRIBUTED SWITCH IN NETWORK 1):

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

Switch(config-if)#
```

STEP 2:

VLAN 30,40:

SW3(DISTRIBUTED SWITCH IN NETWORK 2):

```
Switch>
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

Switch(config-if)#
```

ROUTE 1:

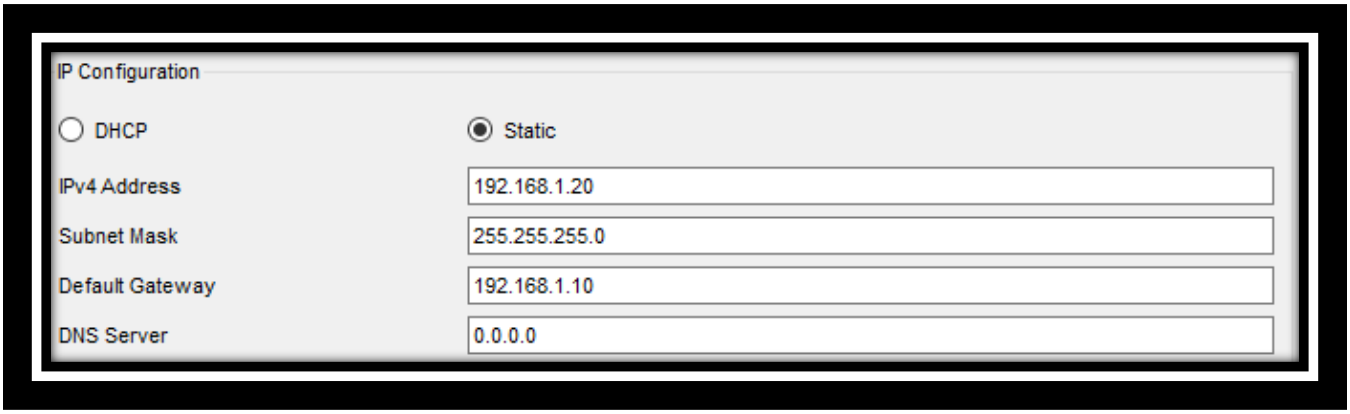
```
Router>
Router>
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface giga0/0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#interface giga0/0/0.30
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.30, changed state to
up

Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip add 192.168.3.10 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface giga0/0/0.40
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.40, changed state to
up

Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip add 192.168.4.10 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#
```

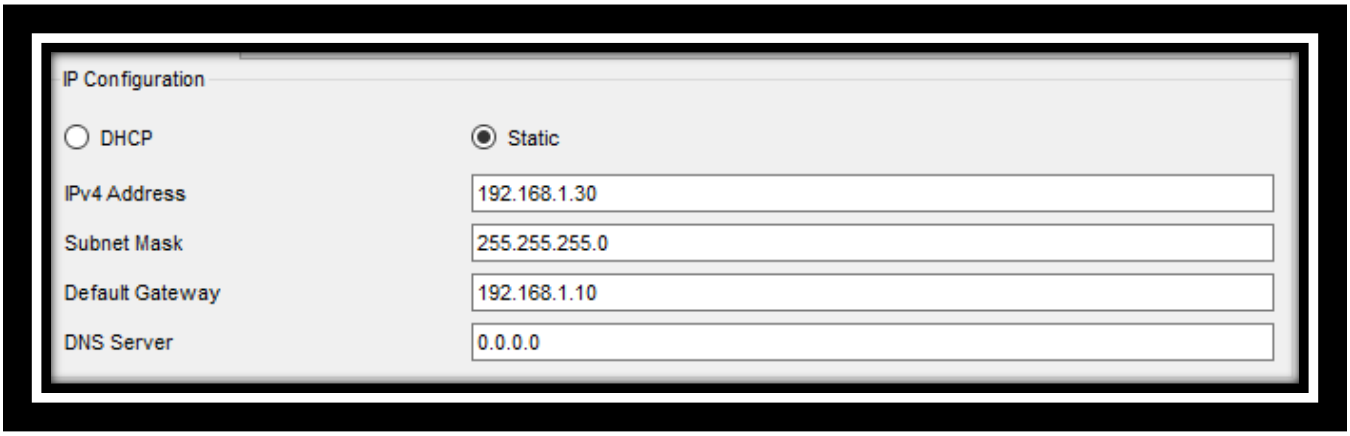
IP CONFIGURATION

STEP 1:

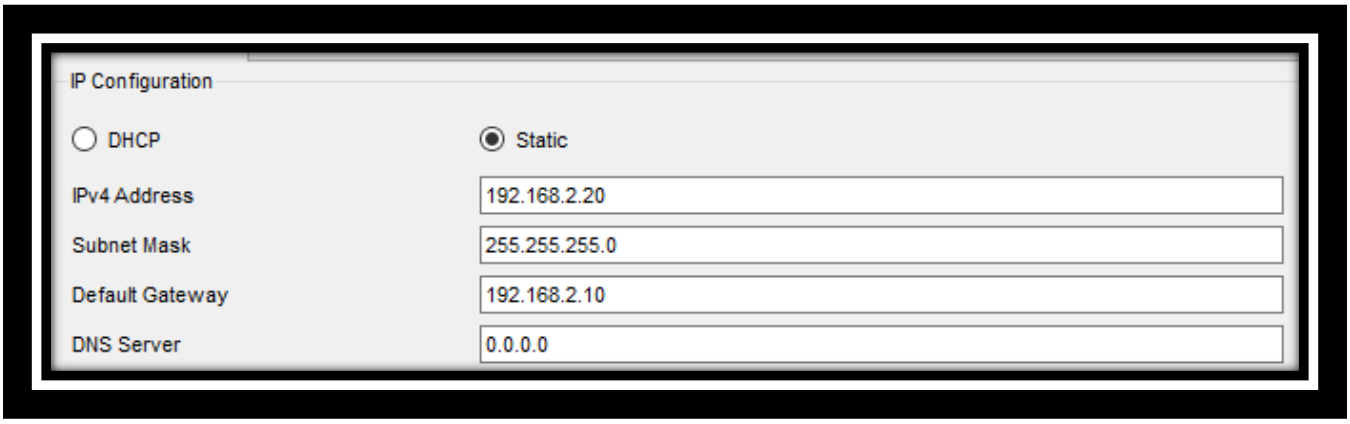
VLAN 10 ,20 :

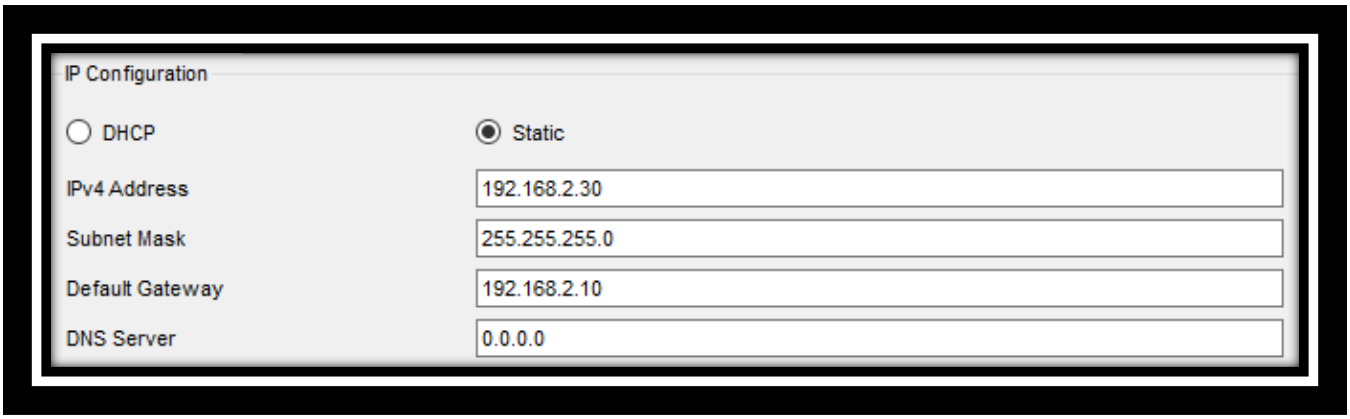
PC 0 ,1:VLAN 10





PC 2,3:VLAN 20

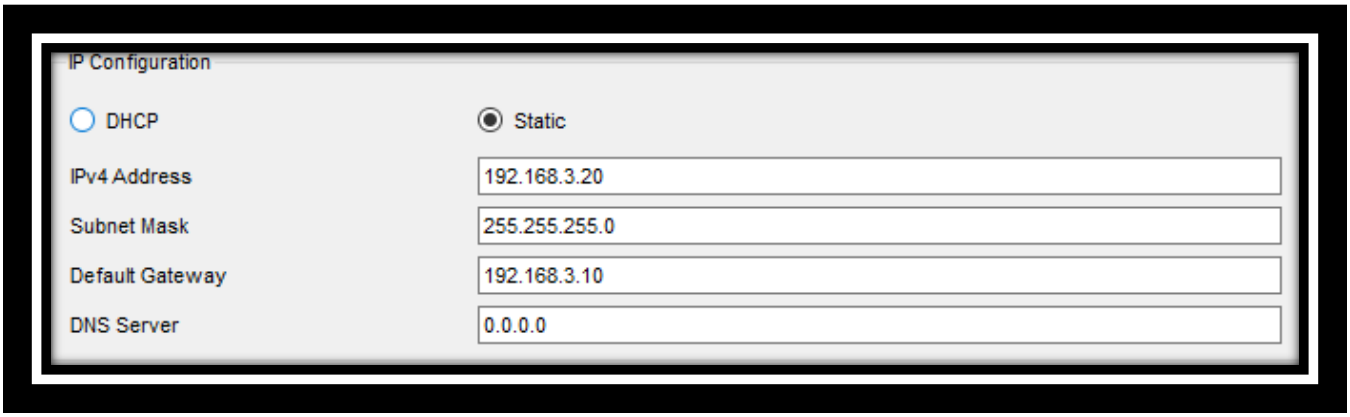


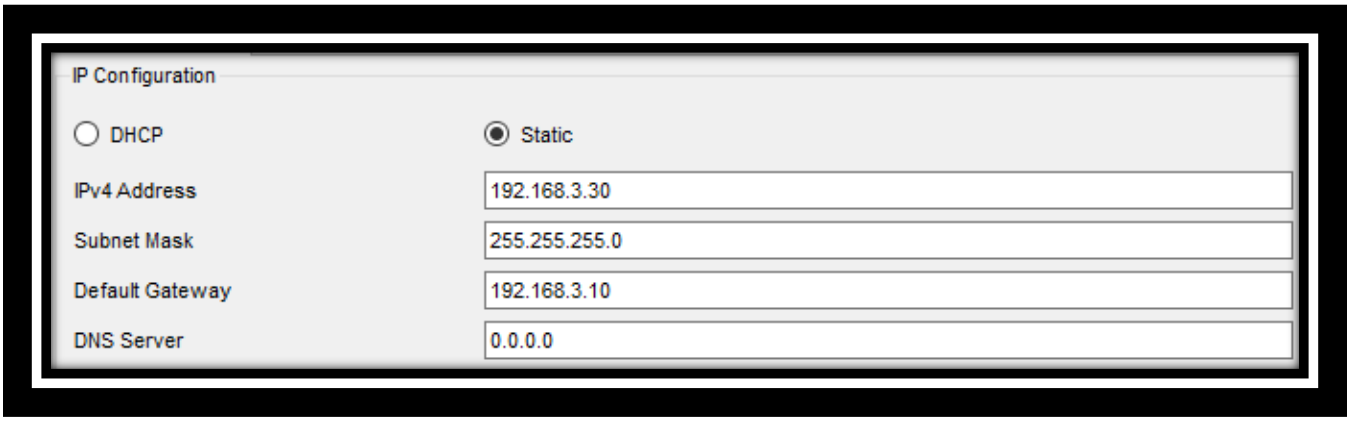


STEP 2:

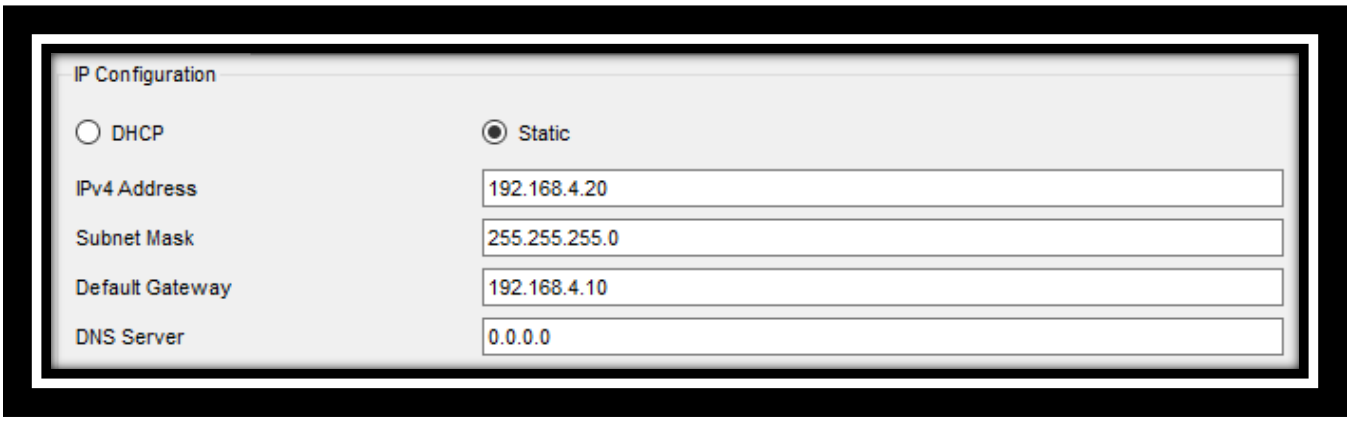
VLAN 30 , 40 :

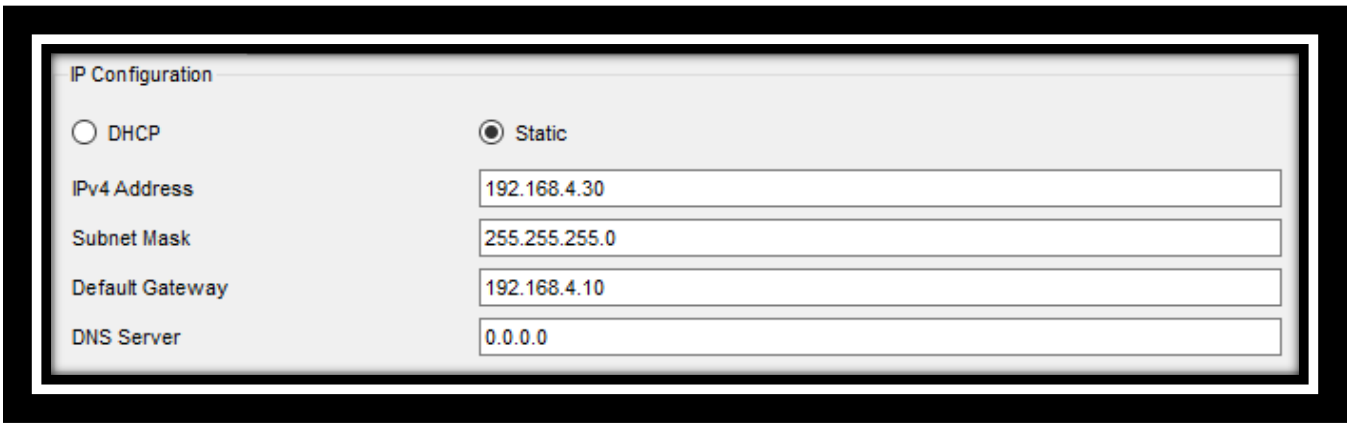
PC 4 , 5:VLAN 30





PC 5,6:VLAN 40


IP Configuration
DHCP
Static
IPv4 Address
192.168.4.20
Subnet Mask
255.255.255.0
Default Gateway
192.168.4.10
DNS Server
0.0.0.0


IP Configuration
DHCP
Static
IPv4 Address
192.168.4.30
Subnet Mask
255.255.255.0
Default Gateway
192.168.4.10
DNS Server
0.0.0.0

STATIC ROUTING

STEP 1:

ROUTE 0:

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface giga0/0/1
Router(config-if)#ip add 10.10.10.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

Router(config)#
Router(config)#ip route 192.168.3.0 255.255.255.0 10.10.10.2
Router(config)#ip route 192.168.4.0 255.255.255.0 10.10.10.2
Router(config)#exit
```

STEP 2:

ROUTE 1:

```
Router>
Router>
Router>en
Router#
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface giga0/0/1
Router(config-if)#ip add 10.10.10.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.10.10.1
Router(config)#ip route 192.168.2.0 255.255.255.0 10.10.10.1
Router(config)#exit
```

OUTPUT :

PING FROM NETWORK 1 TO NEYWORK 2

```
C:\>ping 192.168.3.30

Pinging 192.168.3.30 with 32 bytes of data:

Reply from 192.168.3.30: bytes=32 time<1ms TTL=126
Reply from 192.168.3.30: bytes=32 time<1ms TTL=126
Reply from 192.168.3.30: bytes=32 time=10ms TTL=126
Reply from 192.168.3.30: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
```

PING FORM NETWORK 2 TO NETWORK 3

```
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=126
Reply from 192.168.1.20: bytes=32 time=11ms TTL=126
Reply from 192.168.1.20: bytes=32 time=10ms TTL=126
Reply from 192.168.1.20: bytes=32 time=10ms TTL=126

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 7ms

C:\>
```

ACCESS CONTROL LIST

STEP 1:

VLAN 20 CANNOT ACCESS VLAN 40

VLAN 20 IN NETWORK 1

VLAN 30 IN NETWORK 2

```
Router#
Router#
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 100 deny ip 192.168.2.0 0.0.0.255 192.168.4.0 0.0.0.255
Router(config)#access-list 100 permit ip any any
Router(config)#interface giga0/0/0.20
Router(config-subif)#ip access-group 100 in
Router(config-subif)#
```

OUTPUT:

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.4.20

Pinging 192.168.4.20 with 32 bytes of data:

Reply from 192.168.2.10: Destination host unreachable.
Reply from 192.168.2.10: Destination host unreachable.
Reply from 192.168.2.10: Destination host unreachable.
Reply from 192.168.2.10: Destination host unreachable.

Ping statistics for 192.168.4.20:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```